Power

File: power.kicad_sch

Controller

File: controller.kicad_sch

Sound chip

File: sound_chip.kicad_sch

Audio out

File: audio.kicad_sch

Front panel

File: panel.kicad_sch

How to read this schematic
Each box is a page. Wires end in name tags (labels): any two tags with the same name are connected,
even on different pages. The symbols marked +5V, +5V_YM, +4V5A, +3V3 and GND are power connections.
Parts marked DNP are not fitted by PCBWay. Pages: Power > Controller > Sound chip > Audio out > Front panel.

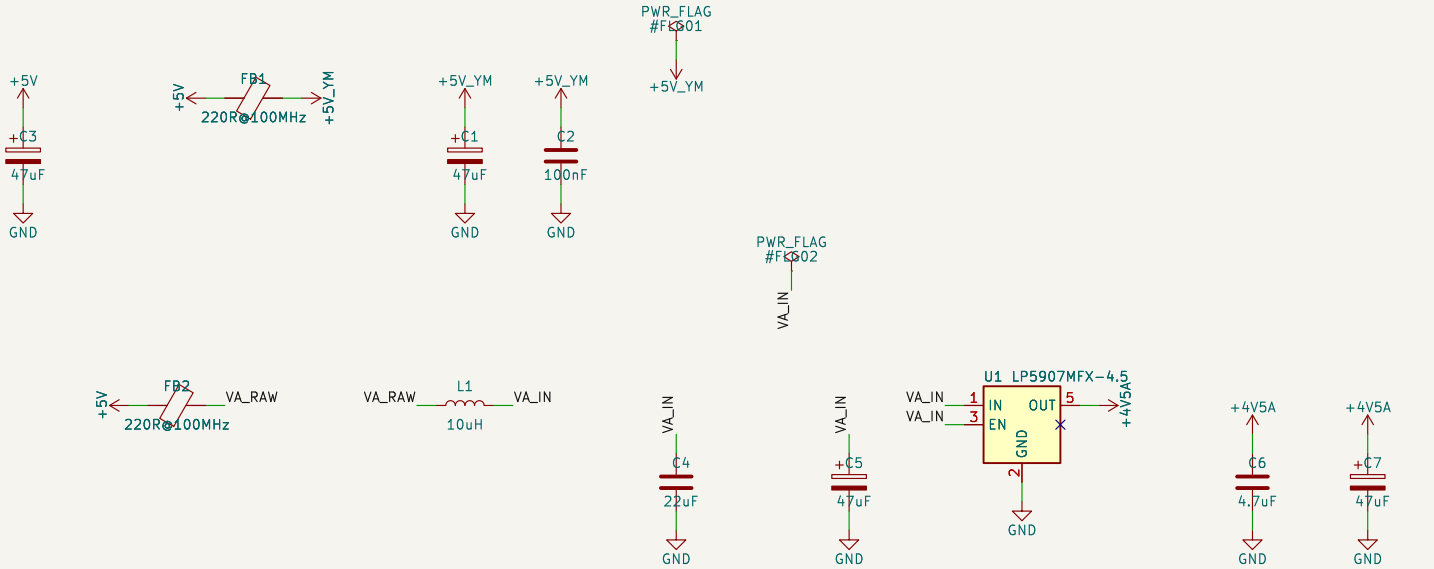
Top sheet: double-click a box to open that page
YM2612

Sheet: /
File: ym2612_player.kicad_sch

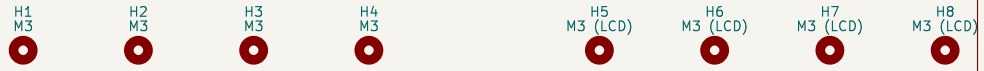
Title: YM2612 player Rev A

Size: A4	Date: 2026-10-24	**Rev: A0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/6

POWER: everything runs from the Pico Plus 2's own USB-C port (VBUS, 5V, up to 500 mA from a computer).
+5V = VBUS from the Pico (pin 40). Feeds the LCD, backlight, window LEDs.
+5V_YM = +5V through a ferrite: YM2612 and the two 74HCT541 buffers.
+4V5A = quiet analog rail: ferrite + 10uH/22uF+47uF filter + LP5907 4.5V low-noise LDO. Op-amps and headphone amp.
+3V3 = made by the regulator on the Pico board.
GND is one solid plane on both layers. The audio parts are grouped on the left, away from USB and the LCD,
so digital return currents don't flow under them. (A split ground put the YM2612 bus over the audio ground.)



Mounting holes: M3, positions set by the enclosure. H5-H8 hold the LCD module's standoffs.

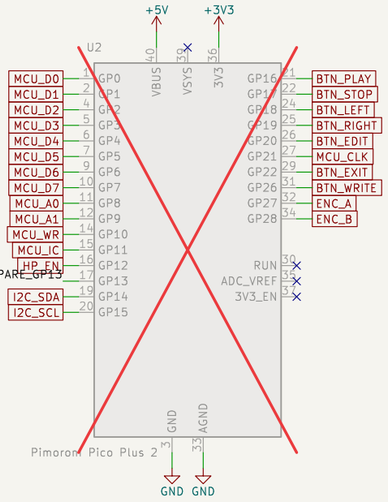


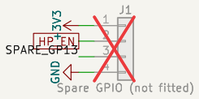
YM2612 player Rev A, draft for review
YM2612
Sheet: /Power/
File: power.kicad_sch

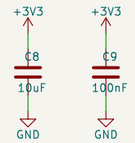
Title: Power

Size: A4	Date: 2026-10-24	**Rev: A0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/6

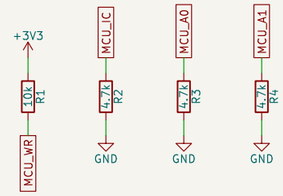
CONTROLLER: Pimoroni Pico Plus 2 (RP2350, 16 MB flash, 8 MB PSRAM, USB-C). It has parts underneath, so it plugs into two 20-pin sockets.
Its USB-C port is the player's only power and data connector: put it on the rear edge of the board.
Pin plan (from docs/DESIGN.md 5.4)
  GP0-GP7 YM2612 data D0-D7 (PIO, must stay in order)   GP8, GP9 A0, A1   GP10 /WR   GP11 /IC (reset)
  GP21   7.67 MHz chip clock (GPOUT0: 153.33 MHz / 20)
  GP14, GP15 I2C1 SDA / SCL to the LCD (through the level shifter on the Front panel page)
  GP16 Play GP17 Stop GP18 Left GP19 Right GP20 Edit GP22 Exit GP26 Write   (buttons to GND, internal pull-ups)
  GP27, GP28 data knob A / B
  GP12   headphone amp enable (high = on), also on J1   GP13 spare, on J1 (not fitted)
/RD is tied high and /CS low at the chip, so the bus only ever writes.







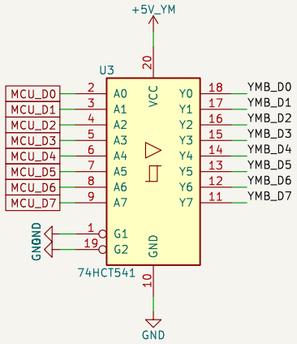
Test points for the logic analyzer (milestone M1)

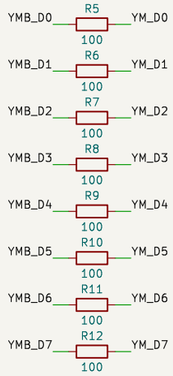


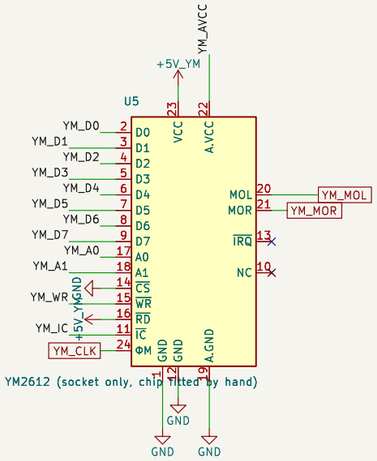
Boot-time pulls: until the firmware starts, these hold the
YM2612 in reset and keep /WR idle. 4.7k pull-downs: RP2350 erratum E9.

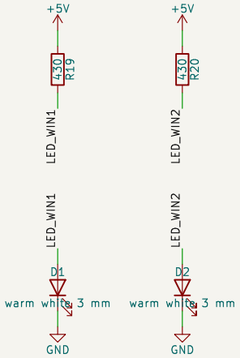
YM2612 player Rev A, draft for review
YM2612
Sheet: /Controller/
File: controller.kicad_sch
Title: Controller
Size: A3 | Date: 2026-10-24 | **Rev: A0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 3/6

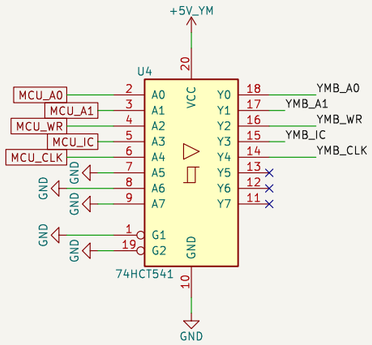
SOUND CHIP: the YM2612 and the 3.3V to 5V buffers
Two 74HCT541 buffers at 5V turn the Pico's 3.3V signals into solid 5V. 100 ohm resistors on every chip pin protect the YM2612.
The bus is write-only: /RD is tied high and /CS low, so the chip listens whenever /WR pulses. /IRQ is not used.
Clock: the Pico makes 7.667 MHz on GP21 and the buffer lifts it to 5V (its 33 ohm resistor is fitted). For exact Genesis pitch,
fit a 7.670454 MHz 5V oscillator can in X1 and its 33 ohm resistor, and leave out the buffer's 33 ohm.
Window lights: D1 and D2 sit beside the socket, angled across the chip, so its markings catch the light.

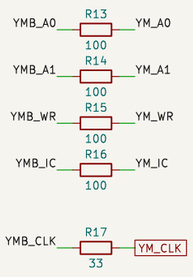


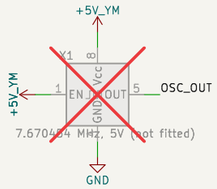




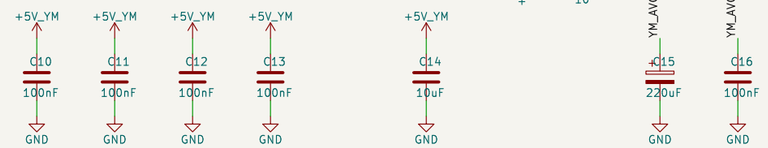












Decoupling: one 100nF at each 74HCT541, one at YM2612 VCC (pin 23) with 10uF bulk, one for X1.
A.VCC (pin 22): 10 ohm + 220uF + 100nF filter, placed at pins 22 and 19.

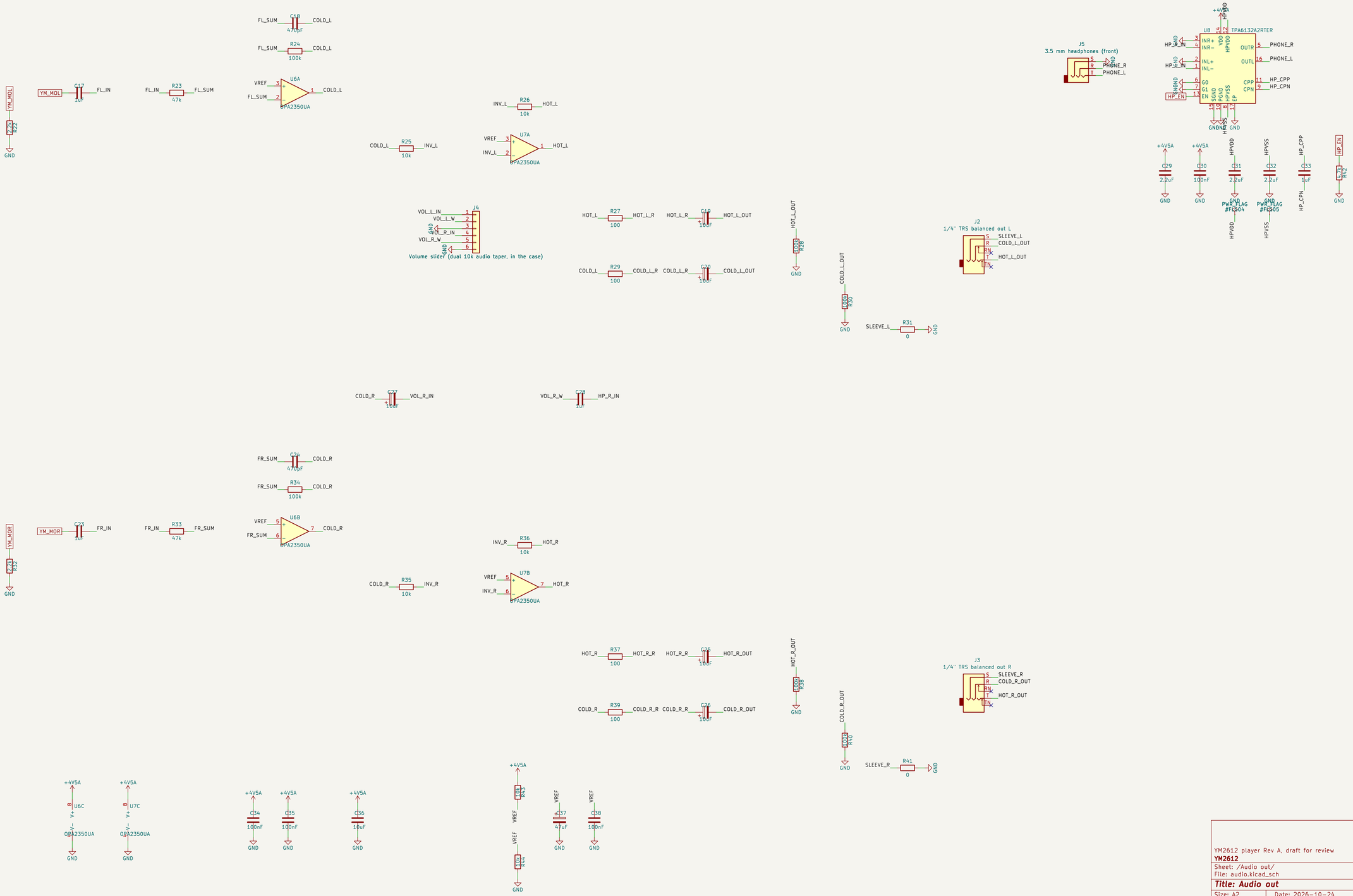
YM2612 player Rev A, draft for review
YM2612

Sheet: /Sound chip/
File: sound_chip.kicad_sch

Title: Sound chip

Size: A3	Date: 2026-10-24	**Rev: A0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/6

AUDIO OUT (authentic Model 1 sound, no clean mode). Everything here runs on the quiet +4V5A rail around VREF = 2.25V.
1. 2.2k load on MOL / MOR, as in the Mega Amp, so the chip's DAC works as it did in the console.
2. Model 1 style low-pass: inverting OPA2350 stage, 47k in, 100k || 470pF feedback. First order, 3.39 kHz, gain about 2.1.
   That matches the 47k / 1nF Model 1 filter used by MIDIbox. Change C to 560pF for 2.84 kHz (the VA3 knee) after the A/B test in M4.
3. Balanced line outs: cold = filter out, hot = filter out inverted again by a 10k / 10k OPA2350 stage. 100 ohm + 10uF + 100k bleed
   on each leg into a Neutrik 1/4in TRS (tip hot, ring cold, sleeve to GND through a 0 ohm ground-lift footprint).
4. Headphones: 10uF DC block, dual slide pot on its 6-pin plug (volume, headphones only), 1uF into a TPA6132A2 at -6 dB gain.
   Its outputs are ground-centred, so they go straight to the front 3.5 mm jack. GP12 (HP_EN) mutes it until the rail settles.



YM2612 player Rev A, draft for review

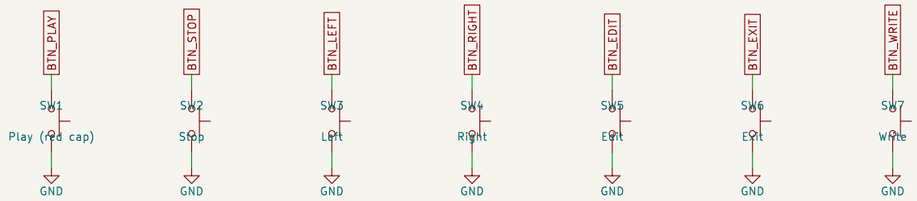
YM2612

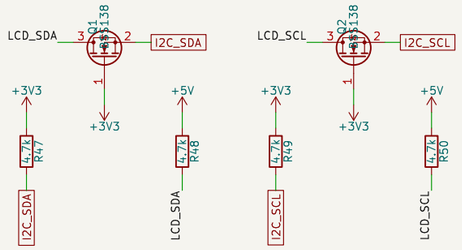
Sheet: /Audio out/
File: audio.kicad_sch

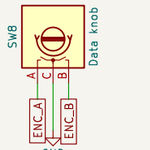
Title: Audio out

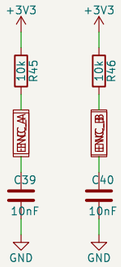
Size: A2	Date: 2026-10-24	**Rev: A0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/6

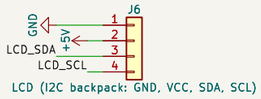
FRONT PANEL: 7 buttons, data knob, LCD connector
Buttons: 12 mm tactile switches with snap-on caps (Joey, Oct 24). Each shorts its Pico pin to GND; the Pico's pull-ups are on.
Data knob: turn-only encoder with 10k pull-ups and 10nF filters for clean steps.
LCD: HiLetgo 20x4 HD44780 blue LCD with a PCF8574 I2C backpack, 5V, address 0x27 (or 0x3F).
The LCD connector's pin order matches the backpack's GND, VCC, SDA, SCL. Make a short 4-wire cable, JST XH to Dupont.
The BSS138 pair shifts I2C between the Pico's 3.3V and the backpack's 5V. Keep the LCD cable away from the audio corner.

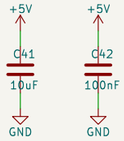












YM2612 player Rev A, draft for review
YM2612
Sheet: /Front panel/
File: panel.kicad_sch
Title: Front panel
Size: A3 | Date: 2026-10-24 | **Rev: A0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 6/6